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Sawaira Mumtaz

SKILLS

AI/ML and Full-Stack Software Engineer with production experience building LLM-powered applications and secure full-stack systems. Proficient in LLM integration (OpenAI, Groq, gemini, OpenRouter, ollama), RAG systems, real-time streaming, React TypeScript, NestJS, MongoDB, JWT authentication, and cloud deployment.

EDUCATION

- **Matriculation in Science (2018)**
Government Girls High School GareebAbad (Rawalpindi)
 - **Intermediate in ICS-Physics (2021)**
Govt. Associate College for Women Dhoke Munshi (Rawalpindi)
 - **Bachelors in Computer Science (Oct 2022 - Mar 2027)**
Virtual University of Pakistan (Saddar Campus Rawalpindi)
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EXPERIENCE

GO Viral Communication, ISB NASTP — Onsite Internship

4 March to 4 June 2026

1. Software Engineering Internship

- Developed Notifications System full-stack application using React TypeScript (frontend), NestJS TypeScript (backend), MongoDB (database), and RESTful APIs
- Implemented secure authentication mechanisms: JWT token-based authentication and bcrypt password hashing for user data protection
- Built comprehensive backend with CRUD operations, data validation, secure API endpoints, and RESTful architecture

- Gained practical experience in modern software development stack: C programming, Python Flask APIs, Git/GitHub, Linux environment setup, Docker containerization, Nginx web server, and full-stack web development
- Demonstrated technical understanding and dedication throughout 12-week engagement with emphasis on frontend design and backend development quality

Serverless Task Manager

Vercel, Supabase, Node.js | Cloud Architecture, Serverless Computing

- Architected full-stack task management application using Vercel serverless functions and Supabase PostgreSQL
- Demonstrated cloud infrastructure optimization and free-tier service selection for scalable, production-ready applications
- Implemented secure API endpoints, database integration patterns, and RESTful architecture

Notifications System

React TypeScript, NestJS TypeScript, MongoDB, JWT, bcrypt, REST APIs

- Designed and developed complete notification platform with React TypeScript frontend and NestJS TypeScript backend
- Implemented secure user authentication (JWT + bcrypt), CRUD operations, and data validation
- Built responsive frontend UI and robust backend APIs for reliable message delivery and user management

- **Post Fusion, Pvt. Ltd. (Pro Fusion AI) -**

E Earn Innovative, Near Marir Hassan RWP

Onsite Internship 1st June to 28 August 2026

2. Artificial Intelligence Engineer Internship – AI & Machine Learning Department

- Developed LinkedIn Post Generator using Python, LangChain, and Ollama, implementing prompt engineering for high-quality content generation and AI automation quality scoring mechanisms
- Designed and deployed AI-powered content generation studio integrating multiple LLM APIs (OpenAI, Groq, OpenRouter) with real-time streaming using React, Node.js, TypeScript, and Server-Sent Events (SSE)

- Built full-stack streaming content platform (React + Node.js/Express) with multi-LLM API integration, vector databases, and real-time response streaming
- Mastered advanced AI/ML concepts: Machine Learning, Deep Learning, Prompt Engineering, LLM Applications, AI Workflows, and AI Agents
- Implemented version control, CI/CD workflows, and collaborative development using Git and GitHub under professional mentorship
- Completed 8-week structured AI/ML curriculum with hands-on exposure to real client projects and industry-standard workflows

RAG Document-Reading Chatbot

Python, LangChain, HuggingFace Embeddings, Chroma, Google Gemini API, Google Colab

- Built Retrieval-Augmented Generation (RAG) chatbot with multi-document ingestion and semantic search capabilities
- Implemented advanced chunking strategies and prompt engineering for optimized answer quality
- Leveraged free-tier APIs and open-source tools for cost-effective deployment

- **Arch Technology, Isb** — *Remote Internships - july to sept 2026*

3. Data Science & Mobile App Development Internships

- Developed a collaborative filtering recommendation system (SVD algorithm) using the Surprise library, achieving industry-standard RMSE scores for rating predictions.
 - Applied K-Means clustering (Scikit-learn) to segment retail customers into 5 distinct personas, enabling targeted marketing strategies.
 - Built cross-platform mobile applications using Flutter and React Native with responsive UIs and seamless backend API integration.
 - Performed data cleaning, visualization, and statistical analysis using Python, Pandas, and NumPy for business intelligence insight
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CORE COMPETENCIES

- **Programming Languages:** Python, JavaScript, TypeScript, Node.js, C
 - **AI/ML:** LLM Integration, Prompt Engineering, RAG Systems, Vector Databases (pgvector, Chroma), Semantic Search, Machine Learning, Deep Learning, AI Automation, AI Agents, Model Testing
 - **Web Development:** React, Express.js, NestJS, Flask, Full-stack Development
 - **Frameworks & Libraries:** LangChain, HuggingFace, TensorFlow.js, Mongoose, JWT Authentication
 - **DevOps & Tools:** Git, GitHub, Docker, Nginx, Linux, GitLab CI/CD, Google Colab
 - **APIs & Integrations:** OpenAI, Groq, OpenRouter, REST APIs, Flask APIs
 - **Database & Authentication:** MongoDB, PostgreSQL, JWT, bcrypt, Secure User Authentication, CRUD Operations
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Language:

- English
- urdu